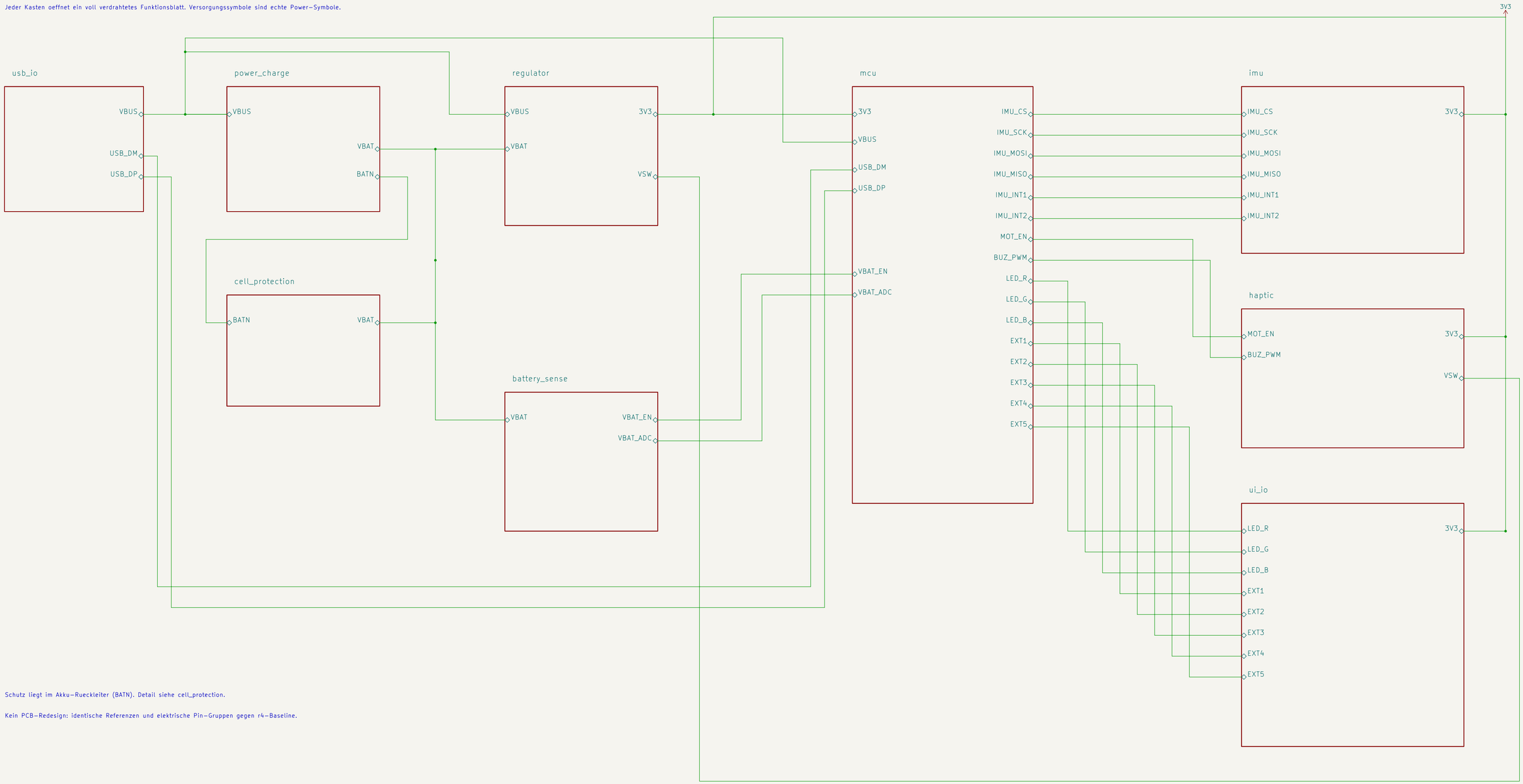
ENERGIE: USB / Akku -> Laden und Schutz -> Regler -> MCU -> Sensorik / Bedienung

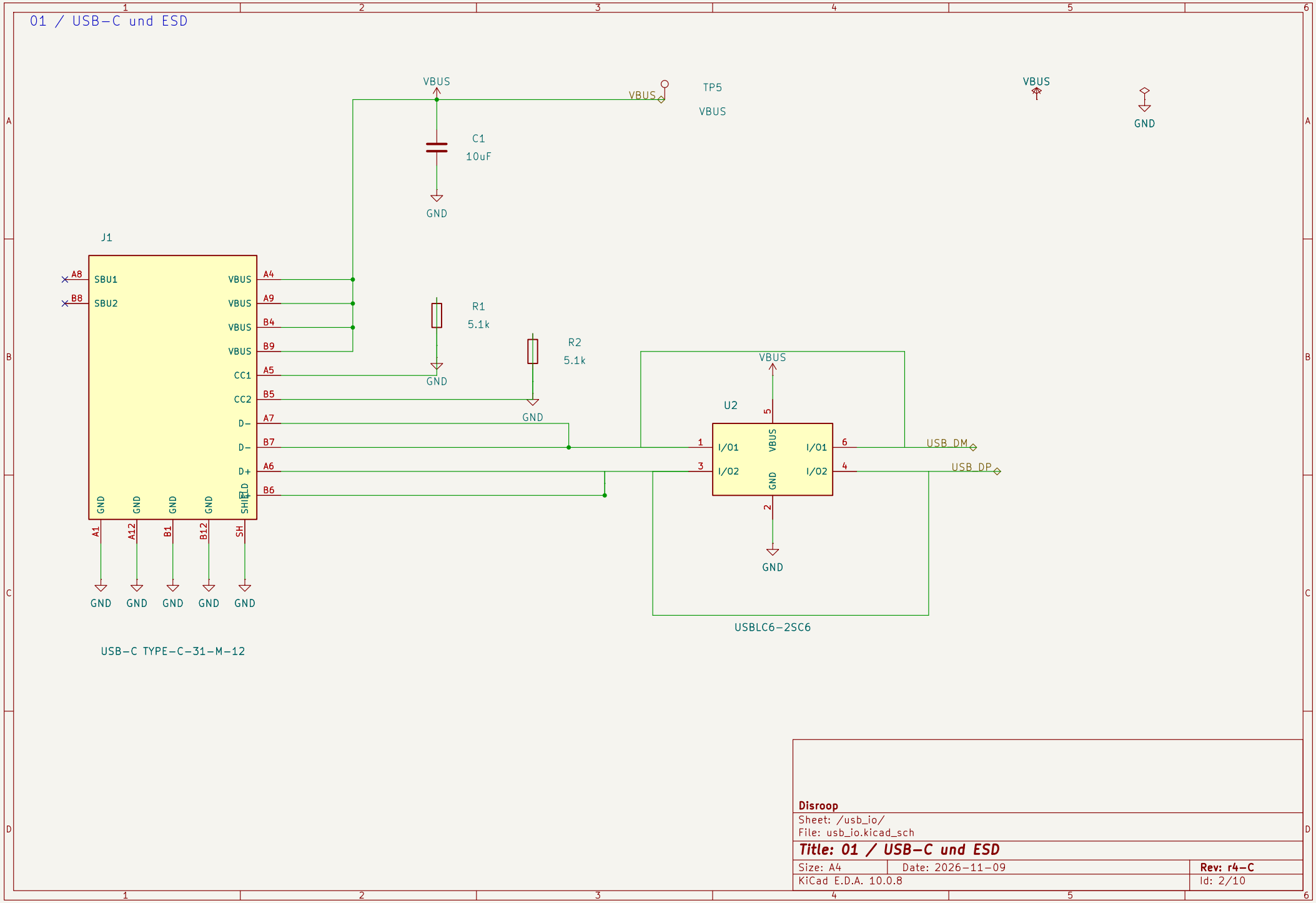
Jeder Kasten oeffnet ein voll verdrahtetes Funktionsblatt. Versorgungssymbole sind echte Power-Symbole.

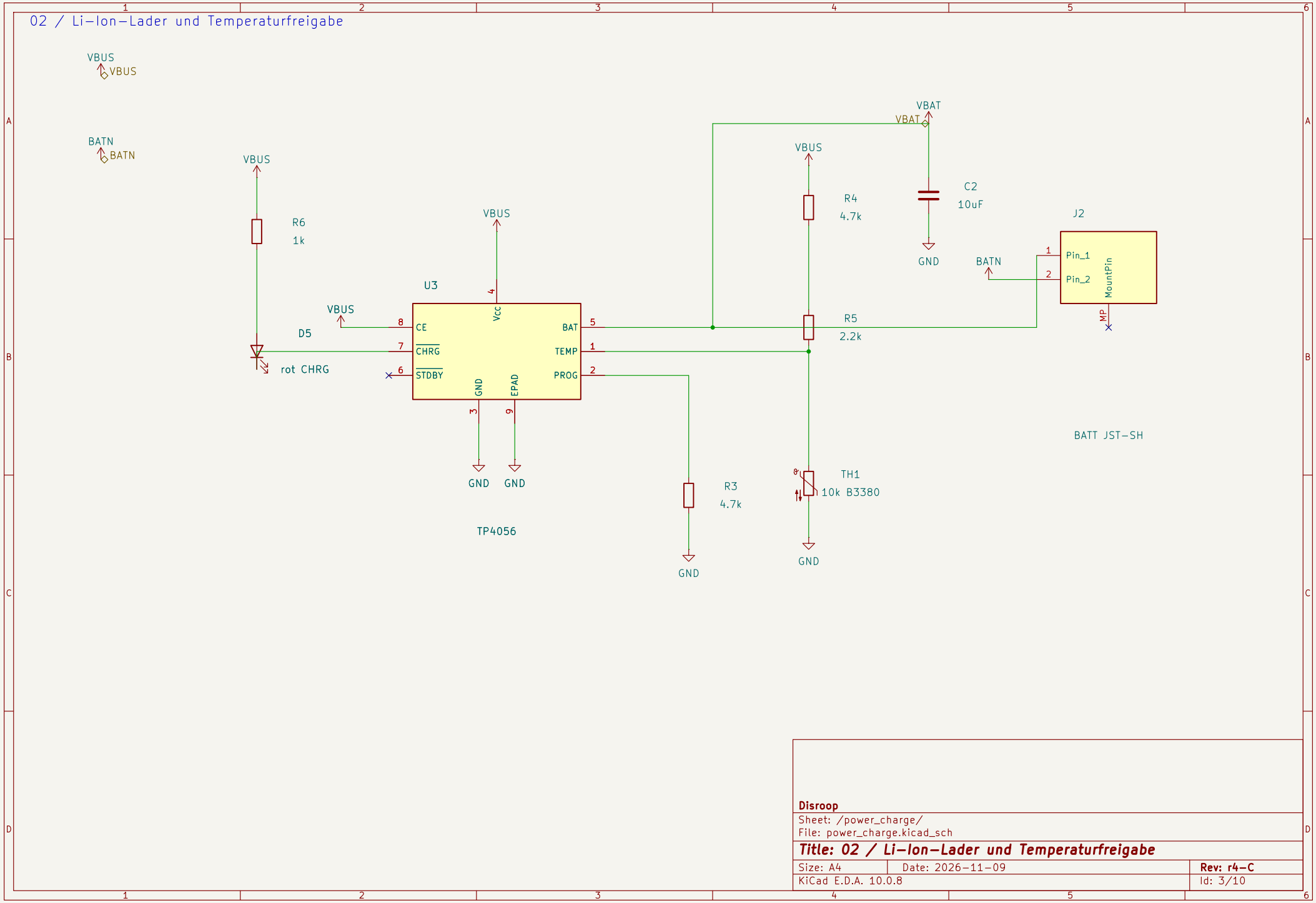


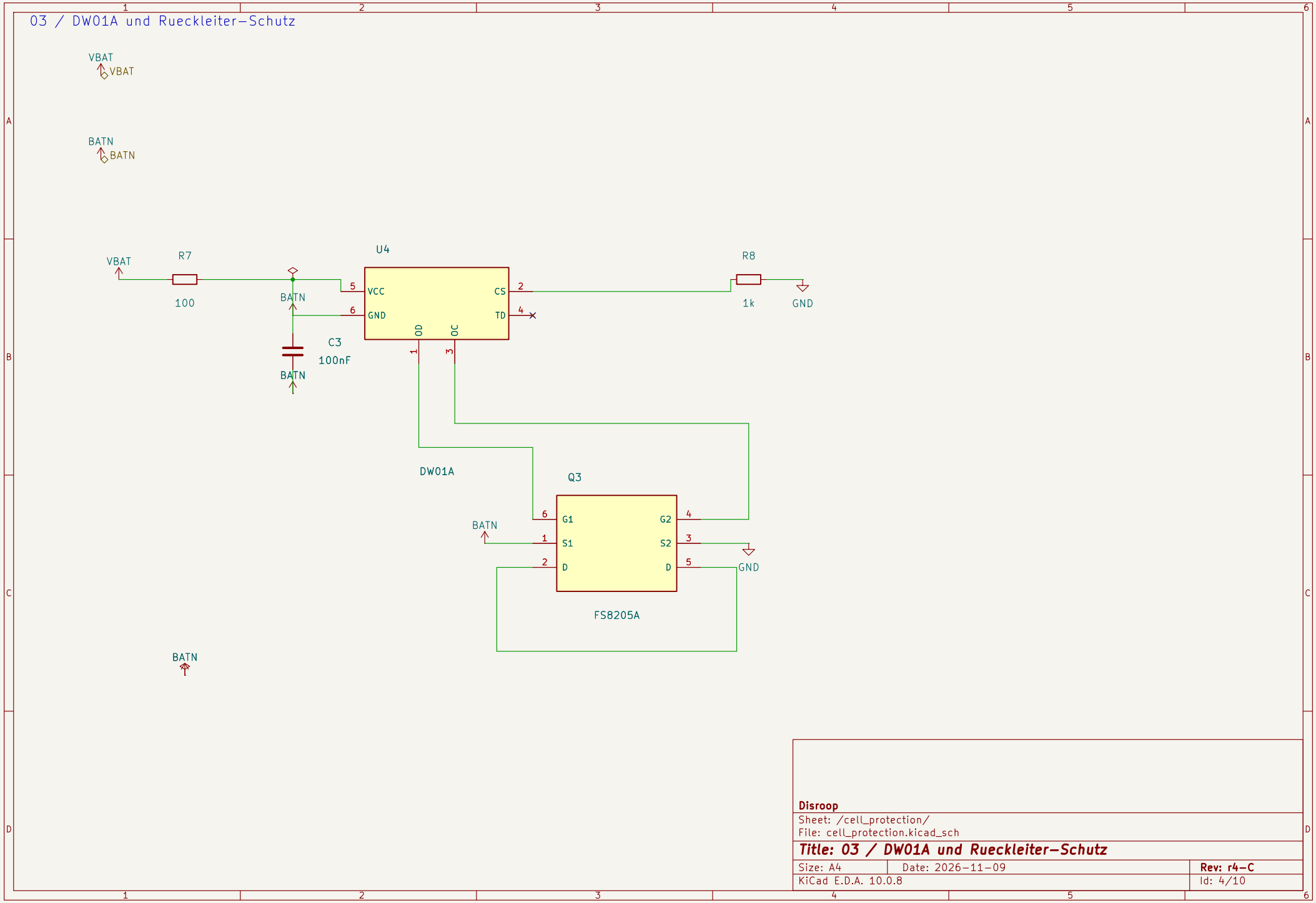
Schutz liegt im Akku-Rueckleiter (BATN). Detail siehe cell_protection.

Kein PCB-Redesign: identische Referenzen und elektrische Pin-Gruppen gegen r4-Baseline.

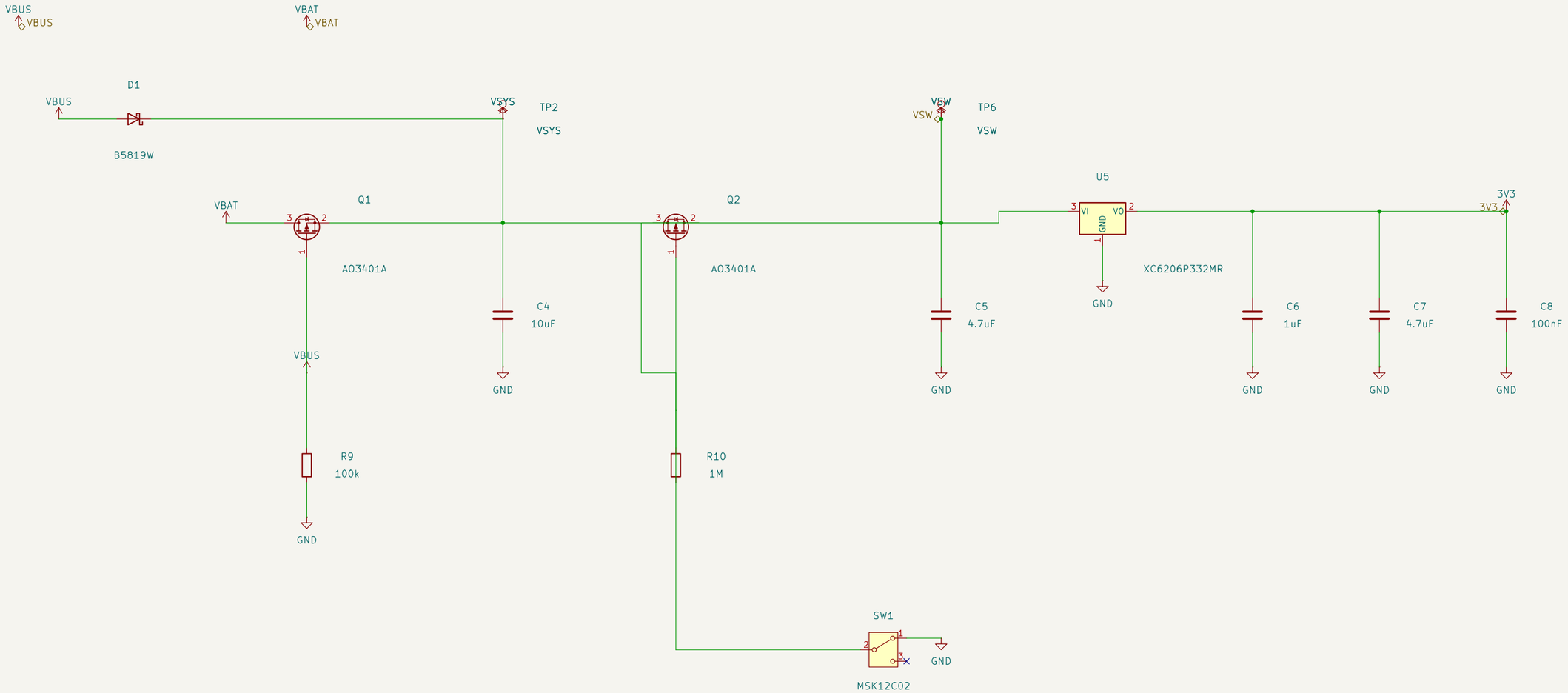
Disroop		
Sheet: /		
File: motion-one.kicad_sch		
Title: Motion One – hierarchische Funktionsarchitektur (Fassung C)		
Size: A2	Date: 2026-11-09	**Rev: r4-C**
KiCad E.D.A. 10.0.8		Id: 1/10


01 / USB-C und ESD
VBUS
TP5
VBUS
C1
10uF
GND
J1
A8
SBU1
B8
SBU2
VBUS
A4
VBUS
A9
VBUS
B4
VBUS
B9
CC1
A5
CC2
B5
D-
A7
D-
B7
D+
A6
B6
SHIELD
GND
A1
A12
B1
B12
SH
R1
5.1k
R2
5.1k
U2
I/O1
I/O2
VBUS
GND
USB_DM
USB_DP
USBLC6-2SC6
USB-C TYPE-C-31-M-12
Disroop
Sheet: /usb_io/
File: usb_io.kicad_sch
Title: 01 / USB-C und ESD
Size: A4
Date: 2026-11-09
Rev: r4-C
KiCad E.D.A. 10.0.8
Id: 2/10

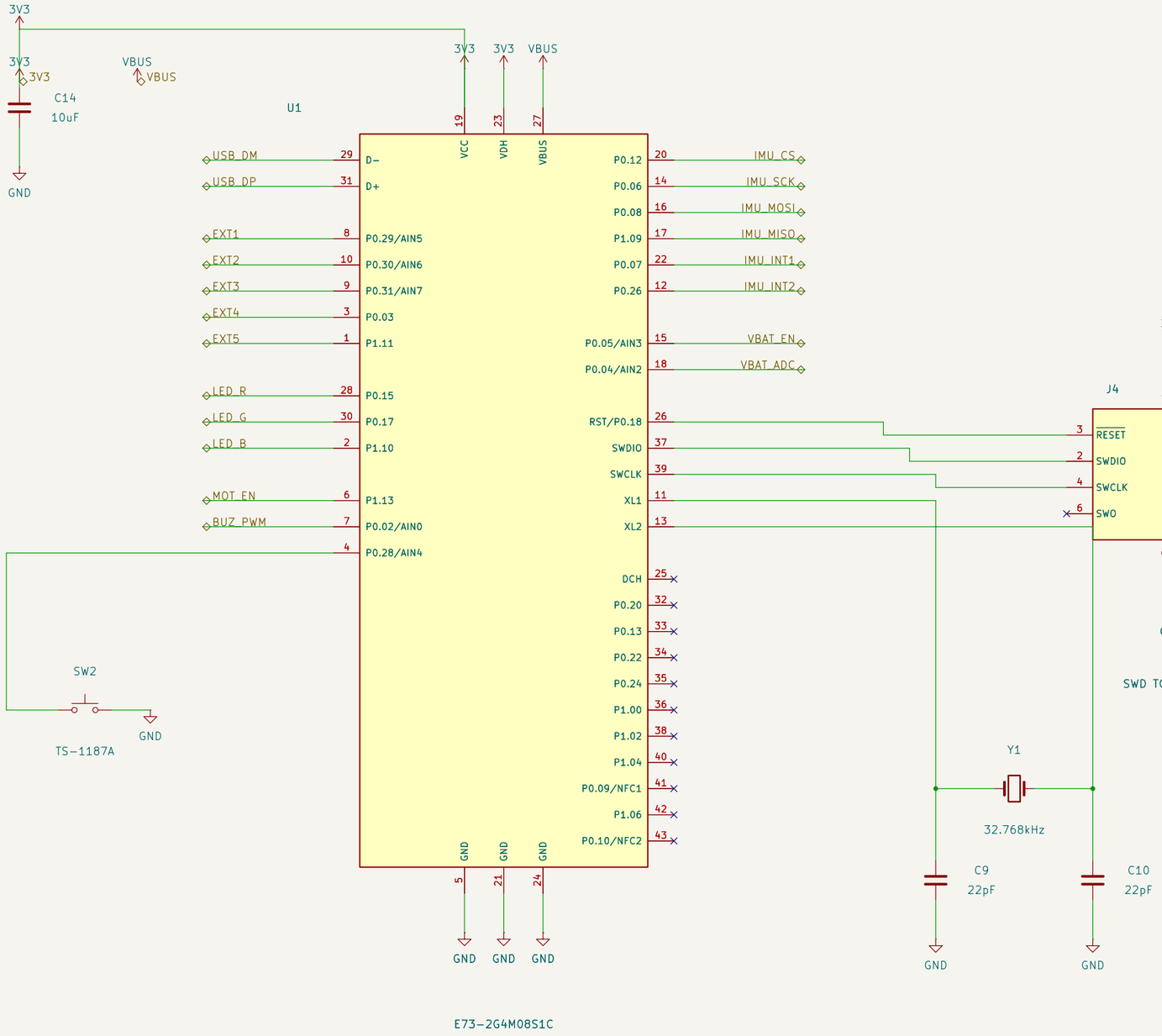

02 / Li-Ion-Lader und Temperaturfreigabe
VBUS
VBUS
BATN
BATN
VBUS
R6
1k
D5
rot CHRG
VBUS
U3
VBUS
4
Vcc
8
CE
7
CHRG
6
STDBY
GND
3
EPAD
9
BAT
5
TEMP
1
PROG
2
GND
GND
TP4056
R3
4.7k
GND
VBUS
R4
4.7k
VBAT
VBAT
C2
10uF
GND
R5
2.2k
TH1
10k B3380
GND
J2
BATN
1
Pin_1
2
Pin_2
MountPin
MP
BATT JST-SH
Disroop
Sheet: /power_charge/
File: power_charge.kicad_sch
Title: 02 / Li-Ion-Lader und Temperaturfreigabe
Size: A4
Date: 2026-11-09
Rev: r4-C
KiCad E.D.A. 10.0.8
Id: 3/10


03 / DW01A und Rueckleiter-Schutz
VBAT
VBAT
BATN
BATN
VBAT
R7
100
BATN
U4
5
VCC
6
GND
OD
OC
1
3
CS
2
TD
4
R8
1k
GND
C3
100nF
BATN
DW01A
Q3
6
G1
1
S1
2
D
G2
4
S2
3
D
5
BATN
GND
FS8205A
BATN
Disroop
Sheet: /cell_protection/
File: cell_protection.kicad_sch
Title: 03 / DW01A und Rueckleiter-Schutz
Size: A4
Date: 2026-11-09
Rev: r4-C
KiCad E.D.A. 10.0.8
Id: 4/10

04 / Power–Path, Hauptschalter und 3V3–Regler

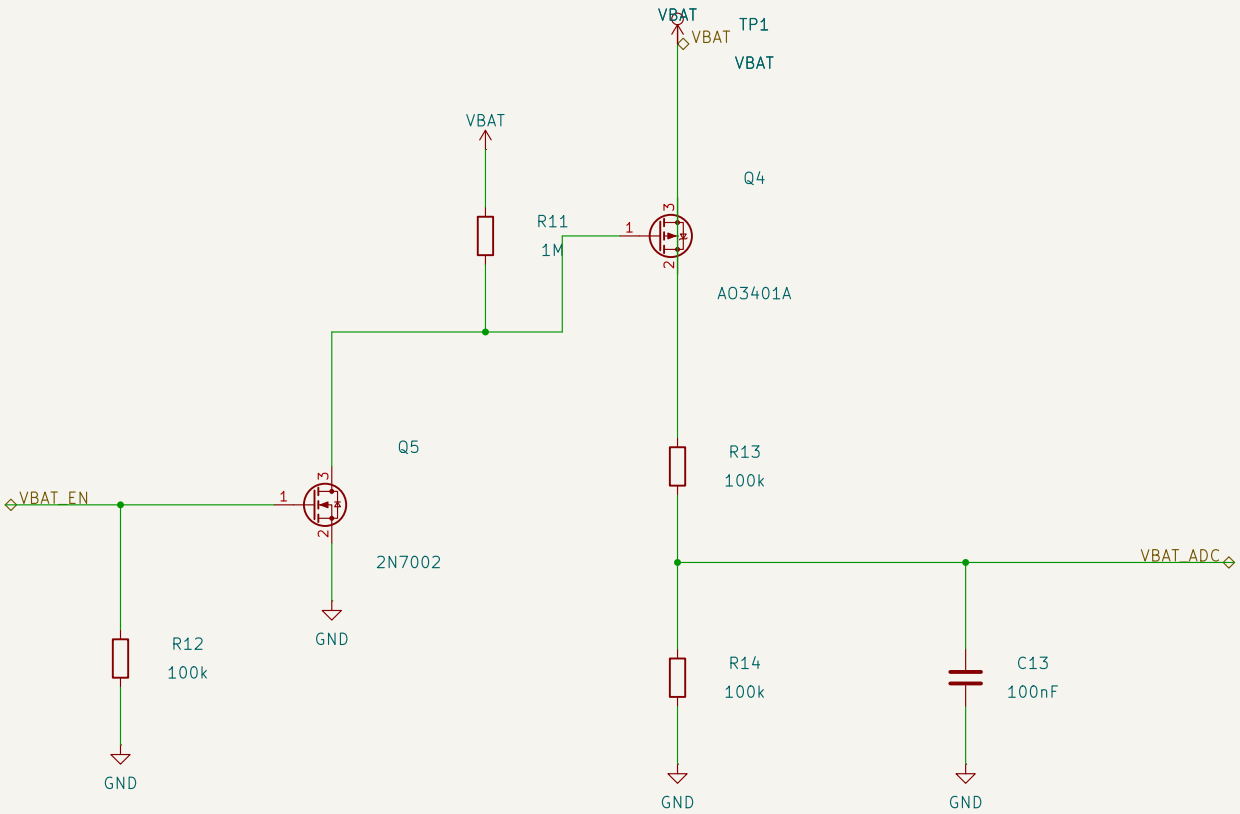


Disroop		
Sheet: /regulator/		
File: regulator.kicad_sch		
Title: 04 / Power–Path, Hauptschalter und 3V3–Regler		
Size: A3	Date: 2026–11–09	**Rev: r4–C**
KiCad E.D.A. 10.0.8		Id: 5/10



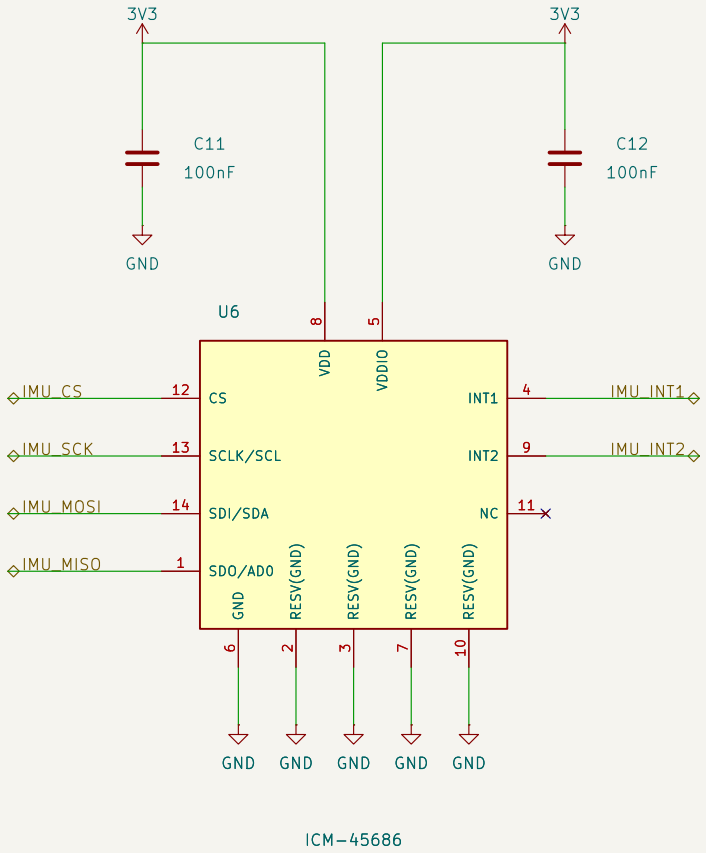
Disroop		
Sheet: /mcu/		
File: mcu.kicad_sch		
Title: 05 / nRF52840, 32-kHz-Quarz, Reset und SWD		
Size: A3	Date: 2026-11-09	**Rev: r4-C**
KiCad E.D.A. 10.0.8		Id: 6/10

06 / Geschalteter Akku-Messpfad



Disroop		
Sheet: /battery_sense/		
File: battery_sense.kicad_sch		
Title: 06 / Geschalteter Akku-Messpfad		
Size: A4	Date: 2026-11-09	**Rev: r4-C**
KiCad E.D.A. 10.0.8		Id: 7/10

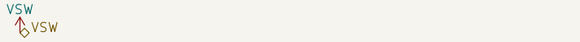
07 / ICM-45686 SPI und Abblockung

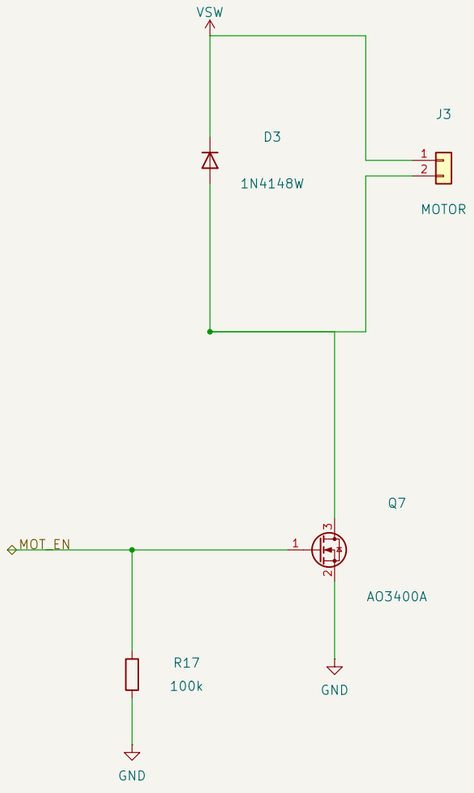


Disroop		
Sheet: /imu/		
File: imu.kicad_sch		
Title: 07 / ICM-45686 SPI und Abblockung		
Size: A4	Date: 2026-11-09	**Rev: r4-C**
KiCad E.D.A. 10.0.8		Id: 8/10

08 / Motor– und Summer–Treiber

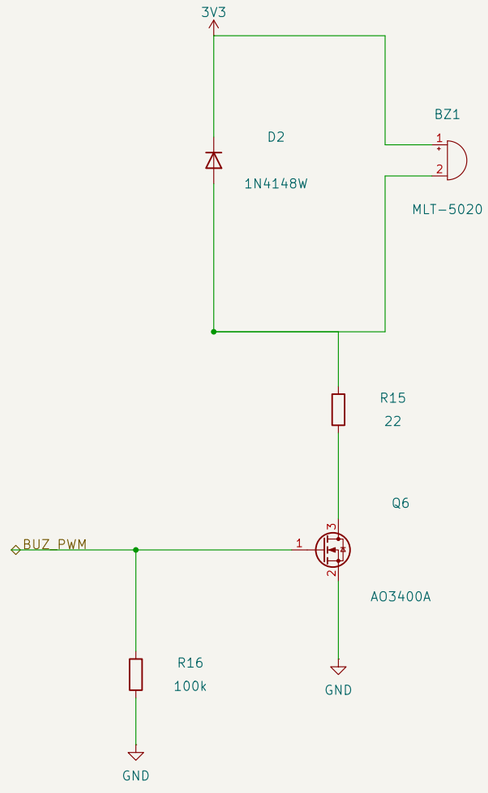
VSW





3V3



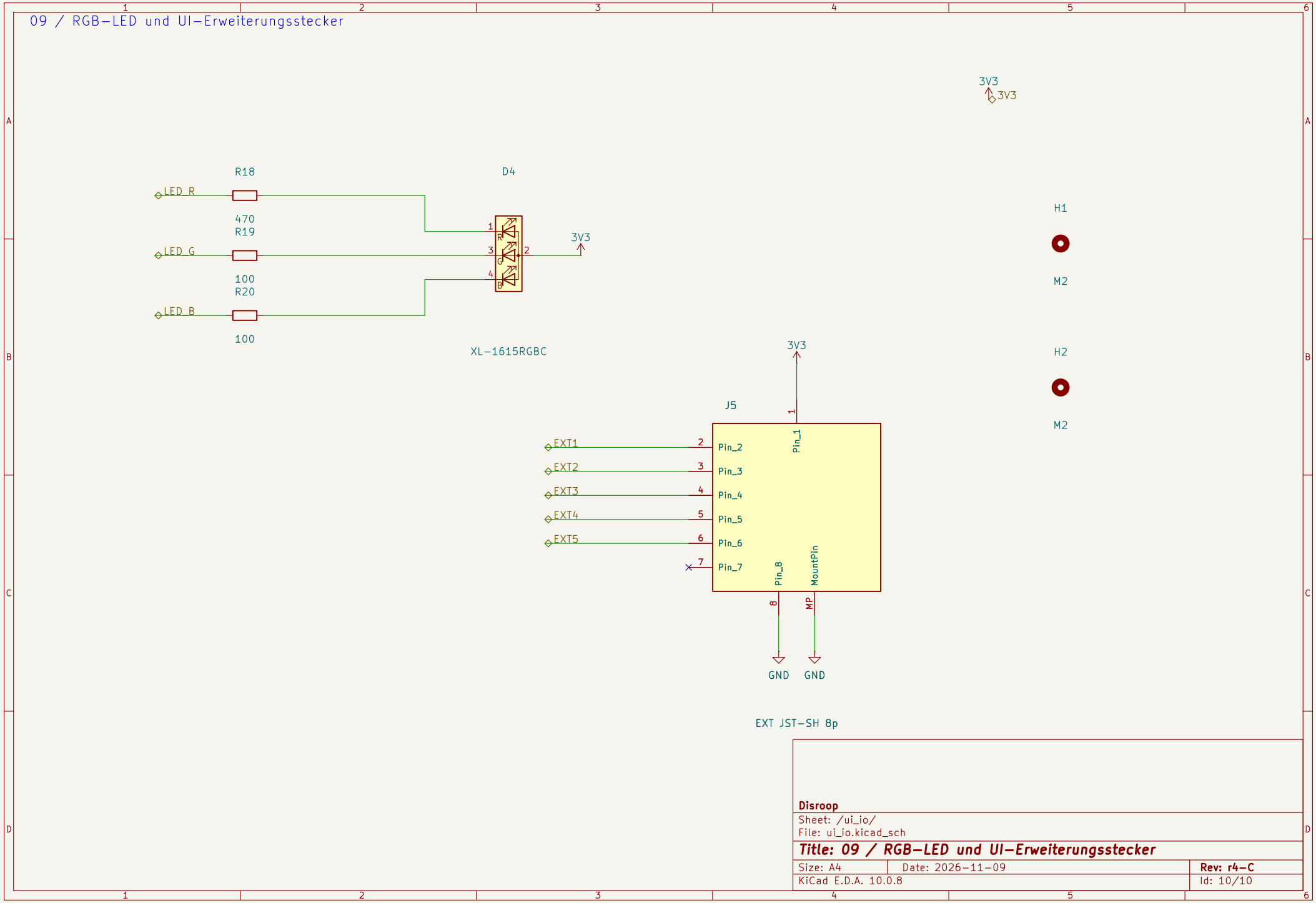


Disroop

Sheet: /haptic/
File: haptic.kicad_sch

Title: 08 / Motor– und Summer–Treiber

Size: A3	Date: 2026–11–09	**Rev: r4–C**
KiCad E.D.A. 10.0.8		Id: 9/10


09 / RGB-LED und UI-Erweiterungsstecker
3V3
3V3
R18
D4
LED_R
470
R19
LED_G
100
R20
LED_B
100
XL-1615RGBC
3V3
H1
M2
H2
M2
3V3
J5
EXT1
EXT2
EXT3
EXT4
EXT5
Pin_1
Pin_2
Pin_3
Pin_4
Pin_5
Pin_6
Pin_7
Pin_8
MountPin
GND
GND
EXT JST-SH 8p
Disroop
Sheet: /ui_io/
File: ui_io.kicad_sch
Title: 09 / RGB-LED und UI-Erweiterungsstecker
Size: A4
Date: 2026-11-09
Rev: r4-C
KiCad E.D.A. 10.0.8
Id: 10/10

Optionale Erweiterungsplatine (J5), nicht Teil des Geräts r4

UI / Joystick, Taster und Touch mit ESD

3V3
3V3
GND
J1
Pin_1
Pin_2
Pin_3
Pin_5
Pin_6
Pin_4
Pin_7
Pin_8
MountPin
GND
GND
EXT JST-SH 8p
SW1
3V3
3V3
A
A
DTA
DTA
a
c
B
B
GND
GND
GND
GND
b
d
GND
GND
GND
GND
GND
GND
GND
GND
RKJXV1224005
R1
100
R2
100
R3
100
STICK_X -> EXT1
STICK_Y -> EXT2
BUTTON -> EXT4
3V3
U2
VP
IO1
IO2
IO3
IO4
VN
GND
SRV05-4
3V3
C1
100nF
GND
H1
M2
H2
M2
E1
Pin_1
TouchPad
R5
1k
U1
VDD
I
Q
VSS
AHLB
TOG
GND
GND
GND
R4
100
TOUCH -> EXT5
TTP233H-BA6

Disroop		
Sheet: /		
File: motion-one-ui.kicad_sch		
Title: Motion One – optionale UI-Platine (Fassung C)		
Size: A3	Date: 2026-11-09	**Rev: r4-C**
KiCad E.D.A. 10.0.8		Id: 1/1